Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Correlation & Regression

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Explain what is meant by positive correlation. **[2]**

2. Explain what is meant by negative correlation. **[2]**

3. State the range of possible values of the product-moment correlation coefficient r. **[1]**

4. Interpret $r = 0.95$. **[2]**

5. Interpret $r = -0.2$. **[2]**

6. Interpret $r = 0$. **[2]**

7. The equation of a regression line is $y = 2.5x + 10$. Interpret the gradient. **[2]**

8. Using $y = 2.5x + 10$, estimate y when $x = 4$. **[2]**

9. Why should you not use a regression line to extrapolate far beyond the data range? **[2]**

10. Given $\sum x = 50$, $\sum y = 80$, $\sum xy = 420$, $\sum x^2 = 300$, $n = 10$, calculate S_{xy} and S_{xx}. **[4]**

11. Using the previous values, find the gradient b of the regression line of y on x. **[2]**

12. Find the equation of the regression line of y on x given $\bar{x} = 5$, $\bar{y} = 8$, $b = 1.2$. **[3]**

13. State which variable is the explanatory (independent) variable in a regression of y on x. **[1]**

14. A scatter diagram shows points tightly clustered about a rising line. Comment on correlation. **[2]**

15. Give one reason why correlation does not imply causation. **[2]**

Section B – Hard (Q16–25)

16. Data: $n = 8$, $\sum x = 40$, $\sum y = 56$, $\sum xy = 310$, $\sum x^2 = 230$, $\sum y^2 = 420$. Calculate r. **[6]**

17. Using the data above, find the equation of the regression line of y on x. **[5]**

18. A regression line is $y = 0.8x + 5$. The mean of x is 10. Find the mean of y. **[2]**

19. Explain the difference between the regression line of y on x and of x on y. **[3]**

20. Given $r = 0.7$, $S_{xx} = 50$, $S_{yy} = 80$, find S_{xy}. **[3]**

21. Residuals are defined as observed y minus predicted y. What does a large residual indicate? **[2]**

22. A data set has $r = 0.92$. After removing one outlier, $r = 0.55$. Comment. **[3]**

23. Calculate the coefficient of determination r^2 when $r = 0.8$ and interpret it. **[3]**

24. For coded data $u = x - 10$, $v = y - 20$, the regression of v on u is $v = 1.5u + 3$. Find the regression of y on x. **[4]**

25. Two variables have $r = -0.95$. Sketch a typical scatter diagram and comment on the strength and direction. **[3]**

End of Worksheet • *Show all working clearly.*